

Investigable Questions & Fair-Test Planning

Year 6 | Australian Curriculum V9.0 | AC9S6I01-AC9S6I02

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Name: _____

Date: _____

LEARNING INTENTION & WORKED MODEL

I can apply scientific inquiry and reasoning to plan, evaluate or communicate evidence. Model: make a clear claim, cite relevant evidence, then explain the link.

A. QUESTION

Turn "plants grow" into a specific investigable question.

B. VARIABLES

For testing light and plant growth, identify changed, measured and controlled variables.

C. HYPOTHESIS

Write an evidence-based prediction.

D. METHOD

Write ordered steps with repeat trials.

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E. SAFETY

Identify two hazards and controls.

F. FAIRNESS

Explain why soil amount must stay the same.

G. SAMPLE

Explain why several plants improve reliability.

H. CHALLENGE

Design a fair test of dissolving rate.

MY LEARNING REVIEW

- ☐ I used evidence or calculations.
- ☐ I explained my reasoning clearly.
- ☐ I checked and improved my work.

The hardest part was: _____

PARENT/CARER TIP

Ask the student to point to the evidence behind every scientific decision.

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ANSWER KEY & TEACHER SUPPORT

A

Example: How does hours of light affect plant height over 14 days?

B

Changed: light; measured: growth; controlled: plant/soil/water/time.

C

Testable prediction with reason.

D

Replicable ordered method.

E

Two relevant hazard-control pairs.

F

It prevents soil amount affecting growth.

G

Reduces influence of individual variation.

H

One changed variable, measured outcome and controls.

ASSESSMENT NOTES

Record whether the student worked independently, used the model, justified decisions and checked accuracy. Re-teach one misconception before extension.